COLLEGE OF COMPUTER STUDIES AND ENGINEERING

Mission

The College of Computer Studies and Engineering aims to develop its students to become useful and responsible citizens utilizing computing and engineering education with well-equipped, accessible, and technologically advanced facilities, and to nurture competent and committed faculty members and staff.

Vision

The College of Computer Studies and Engineering as one of the pillars of Jose Rizal University will lead in pursuit of producing graduates of social importance guided by JRU core values through the use of computing and engineering technology for innovations in teaching and learning.

COMPUTER ENGINEERING PROGRAM

The Bachelor of Science in Computer Engineering (BSCpE) is a program that embodies the science and technology of design, development, implementation, maintenance, and integration of software and hardware components in modern computing systems and computer-controlled equipment.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

Three to five years after completing the Bachelor of Science in Computer Engineering (BSCpE), the graduates will:

1. excel in their profession/career utilizing the knowledge acquired in the Computer Engineering program;
2. become effective collaborators and innovators in the field of computer engineering, applying professional/technical skills and competencies to make a positive impact on society; and
3. be engaged in life-long learning and professional development through self-study, continuing education, or graduate and professional studies.

STUDENT OUTCOMES (SO)

After the completion of the program, the graduates of the BSCpE program shall be able to:

a. apply knowledge of mathematics, sciences, and engineering sciences to the practice of computer engineering.
b. design and conduct experiments as well as analyze and interpret data.
c. design a system, component, or process to meet needs within realistic constraints.

d. work effectively in multi-disciplinary and multi-cultural teams.
e. identify, formulate, and solve computer engineering problems.
f. understand professional and ethical responsibility.
g. communicate effectively.
h. understand the impact of engineering solutions in a global/societal context.
i. engage in life-long learning and to keep current of the development in a specific field of specialization.
j. know contemporary issues.
k. use appropriate techniques, skills, and modern tools necessary for computer engineering practice.
l. know and understand engineering and management principles as a member and leader in a team, to manage projects and in multidisciplinary environments.

CURRICULUM REQUIREMENTS BY YEAR FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER ENGINEERING

FIRST YEAR

First Semester			Units	Second Semester			Units
CPE	C101	Computer Eng'g as a Discipline	1	SSC	C102	Contemporary World	3
HIS	C101	Readings in Philippine History	3	PHI	C102	Ethics	3
ENG	C101	Purposive Communication 1	3	HUM	C102	Art Appreciation	3
FOS	C101	Strategies for Acad Succ in Col	3	MAT	C602	Calculus 2	3
MAT	C101	Mathematics in the Modern World	3	NSC	C104	Physics for Engineers	4
MAT	C601	Calculus 1	3	EGR	C103	Engineering Data Analysis	3
NSC	C204	Chemistry for Engineers	4	CPE	C102	Object-Oriented Programming for CPE	2
CPE	C103	Programming Logic & Design	2	NST	C102	National Service Training Prog 2	3
NST	C101	National Service Training Prog 1	3	PED	C102	(PATHFIT 2)	2
PED	C101	(PATHFIT 1)	2				
			27				26

SECOND YEAR

First Semester			Units	Second Semester			Units
ITC	C201	Database Fundamentals	3	MAT	C202	Linear Algebra	3
MAT	C604	Discrete Structures	3	ENG	C102	Purposive Communication 2	3
NSC	C202	Science, Technology & Society)	3	MAT	C603	Numerical Methods	3
EGR	C101	Computer Aided Drafting	1	CPE	C204	Computer Systems& Troubleshooting	3
EGR	C201	Differential Equations	3	PSY	C101	Understanding the Self	3
EGR	C205	Circuits 1	4	CPE	C315	Computer Application	1
EGR	C203	Engineering Economics	3	ECE	C203	Elec 1: Elec Devices& Circuit	4
CPE	C201	Data Structure & Algorithms	2	CPE	C202	Software Design	4
PED	C201	(PATHFIT 3)	2	PED	C202	(PATHFIT 4)	2
			24				26

THIRD YEAR

First Semester			Units	Second Semester			Units
CPE	C301	Logic Circuits & Design	4	CPE	C313	Artificial Intelligence 1	3
CPE	C303	Data& Digital Communications	3	EGR	C204	Technopreneurship 101	3
CPE	C311	Operating Systems	3	CPE	C302	Computer Networks & Security	4
CPE	C309	Introduction to HDL	1	CPE	C304	Basic Occupational Health & Safety	3
CPE	C307	Cognate/Track Course 1	3	CPE	C306	Microprocessors	4
CPE	C203	Computer Engineering Drafting & Design	1	CPE	C308	Cognate/Track Course 2	3
CPE	C205	Fundamentals of Mixed Signals & Sensors	3	CPE	C305	Digital Signal Processing	4
CPE	C206	Feedback& Control Systems	3				
HIS	C301	Life& Works of Rizal	3				
CPE	C310	CpE Laws & Professional Practice	2				
			26				24

Summer	Units
EGR C302 Methods of Research	3
CPE C401 Computer Architecture & Org	4
	7

Fourth Year

First Semester			Units	Second Semester			Units
CPE	C314	Artificial Intelligence 2	3	CPE	C402	Internship	3
CPE	C405	Emerging Technologies in CpE	3	CPE	C404	CpE Practice & Design 2	2
CPE	C403	CpE Practice & Design 1	1				
CPE	C407	Cognate/Track Course 3	3				
CPE	C409	Seminars& Fieldtrips	2				
CPE	C312	Embedded Systems	4				
			16				5

*EGR C105 (Mathematics for Engineering) 5 units as an additional bridging course for the Non-STEM track.

Cognates/Track Courses

Track 1:	Software Development		9
	Advanced Database Management	3	
	Application Development	3	
	Data Science	3	
Track 2:	System and Network Administration		9
	Network and System Administration	3	
	Network Security	3	
	Cyber Security	3	
Track 3:	Information Technology		9
	Online Technology	3	
	Management & Information System	3	
	Emerging Technology	3	

GROUP REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER ENGINEERING (BSCpE)

		Units	Units
1.	Technical Courses		
	a. Mathematics		12
	Calculus 1	3	
	Calculus 2	3	
	Engineering Data Analysis	3	
	Differential Equations	3	
	b. Physical Science		8
	Physics for Engineers	4	
	Chemistry for Engineers	4	
	c. Basic Engineering Science		9
	Computer-Aided Drafting	1	
	Engineering Economics	3	
	Technopreneurship 101	3	
	Engineering Management	2	
	d. Allied Courses		8
	Circuits 1	4	
	Elec 1: Elec Devices& Circuit	4	
	e. Professional Courses		
	Core Courses		85
	Computer Engineering as a Discipline	1	
	Object-Oriented Programming for CPE	2	
	Programming Logic & Design	2	
	Data Structure & Algorithms	2	
	Software Design	4	
	Computer Engineering Drafting & Design	1	
	Computer Systems& Troubleshooting	3	
	Fundamentals of Mixed Signals & Sensors	3	
	Feedback & Control Systems	3	
	Logic Circuits & Design	4	
	Computer Networks & Security	4	
	Basic Occupational Health & Safety	3	
	Data & Digital Communications	3	
	Digital Signal Processing	4	
	Microprocessors	4	
	Introduction to HDL	1	
	CpE Laws & Professional Practice	2	
	Operating Systems	3	
	Embedded Systems	4	
	Artificial Intelligence 1	3	
	Artificial Intelligence 2	3	
	Computer Application	1	
	Computer Architecture & Organization	2	
	Internship	3	

	CpE Practice & Design 1	1	
	CpE Practice & Design 2	2	
	Emerging Technologies in CpE	3	
	Seminars & Fieldtrips	2	
	Methods of Research	3	
	Database Fundamentals	3	
	Numerical Methods	3	
	Discrete Structures	3	
	f. Technical Elective Courses		9
	Cognate/TrackCourse1	3	
	Cognate/TrackCourse2	3	
	Cognate/TrackCourse3	3	
2.	Non-Technical Courses		
	a. General Education Courses		24
	Art Appreciation	3	
	Science, Technology & Society	3	
	Contemporary World	3	
	Readings in Philippine History	3	
	Understanding the Self	3	
	Purposive Communication 1	3	
	Mathematics in the Modern World	3	
	Ethics	3	
	b. GE Electives/Mandated Courses		12
	Life & Works of Rizal	3	
	Strategies for Academic Success in College	3	
	Purposive Communication 2	3	
	Linear Algebra	3	
	c. Physical Education		8
	PATHFIT 1	2	
	PATHFIT 2	2	
	PATHFIT 3	2	
	PATHFIT 4	2	
	d. National Service Training Program		6
	National Service Training Program 1	3	
	National Service Training Program 2	3	
		Total	**181**

ELECTRONICS ENGINEERING PROGRAM

Electronics Engineering is a branch of engineering that integrates available and emerging technologies with knowledge of mathematics, natural, social and applied sciences to conceptualize, design, and implement new, improved, or innovative electronic, computer, and communication systems, devices, goods, services, and processes.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

The graduates of the Bachelor of Science in Electronics Engineering (BSEcE) are able to:

1. excel in their profession/career utilizing the knowledge acquired in the Electronics Engineering program;
2. become effective collaborators and innovators in the field of Electronics Engineering, applying professional/technical skills and competencies to make a positive impact on society; and
3. be engaged in life-long learning and professional development through self-study, continuing education, or graduate and professional studies.

STUDENT OUTCOMES (SO)

After the completion of the program, the graduates of the BSEcE program shall be able to:

a. apply knowledge of mathematics, sciences, and engineering sciences to the practice of electronics engineering.
b. design and conduct experiments as well as analyze and interpret data.
c. design a system, component, or process to meet needs within realistic constraints.
d. work effectively in a multi-disciplinary and multi-cultural team.
e. identify, formulate, and solve electronics engineering problems.
f. understand professional and ethical responsibility.
g. communicate effectively.
h. understand the impact of engineering solutions in a global/societal context.
i. engage in life-long learning and to keep current of the development in a specific field of specialization.
j. know contemporary issues.
k. use appropriate techniques, skills, and modern tools necessary for electronics engineering practice.
l. know and understand engineering and management principles as a member and leader in a team, to manage projects and in multidisciplinary environments.
m. understand at least one specialized field of electronics engineering field.

CURRICULUM REQUIREMENTS BY YEAR FOR THE DEGREE OF
Bachelor of Science in Electronics Engineering

FIRST YEAR

First Semester			Units	Second Semester			Units
ECE	C101	Electronics Engineering as a Discipline	1	HUM	C102	Art Appreciation	3
EGR	C101	Computer Aided Drafting	1	PHI	C102	Ethics	3
ENG	C101	Purposive Communication 1	3	EGR	C203	Engineering Economics	3
HIS	C101	Readings in Philippine History	3	MAT	C601	Calculus 1	3
EGR	C103	Engineering Data Analysis	3	NSC	C202	Science Technology & Society	3
EGR	C106	Mathematics for Engineering 1	3	ECE	C102	Material Science & Engineering	3
EGR	C107	Mathematics for Engineering 2	3	NSC	C204	Chemistry for Engineers	4
MAT	C101	Math in the Modern World	3	NST	C102	National Service Training Program 2	3
FOS	C101	Strategies for Acad Succ in Col	3	PED	C102	(PATHFIT 2)	2
NST	C101	Nat'l Service Training Program 1	3	SSC	C102	Contemporary World	3
PED	C101	(PATHFIT 1)	2				
			28				30

Summer			Units
ECE	C201	Computer Programming	2
MAT	C602	Calculus 2	3
			7

SECOND YEAR

First Semester			Units	Second Semester			Units
ECE	C104	Physics for ECE	4	ECE	C202	Electromagnetics	4
ECE	C203	Electronic 1: Electronics Devices & Circuit	4	ECE	C204	Electronics 2: Electronic Circuit Analysis & Design	4
ECE	C207	Electronics Shopworks	1	ECE	C206	Communications 1: Principles of Communication Systems	4
EGR	C201	Differential Equations	3	ECE	C215	Integrated Course 1	1
EGR	C205	Circuits 1	4	EGR	C206	Circuits 2	4
NSC	C104	Physics for Engineers	4	ENG	C102	Purposive Communication 2	3
PED	C201	(PATHFIT 3)	2	HIS	C301	Life& Works of Rizal	
PSY	C101	Understanding the Self	3	PED	C202	(PATHFIT 4)	2
			25				22

THIRD YEAR

First Semester			Units	Second Semester			Units
ECE	C205	Advanced Engineering Mathematics	4	ECE	C302	Digital Electronics 2: Microprocessor & Microcontroller Systems	4
ECE	C301	Digital Electronics 1: Logic Circuits & Switching Theory	4	ECE	C306	Communications 3: Transmission Media & Antenna System Design	4
ECE	C303	Electronics 3: Electronics Systems & Design	4	ECE	C308	Communications 4: Data Communications	4
ECE	C305	Communications 2: Modulation & Coding Techniques	4	ECE	C314	Methods of Research	3
ECE	C307	Signals, Spectra, Signal Processing	4	ECE	C315	Integrated Course 2	2
MAT	C202	Linear Algebra	3	EGR	C204	Technopreneurship 101	3
			23				24

Summer			Units
ECE	C309	Internship	3
			3

FOURTH YEAR

First Semester		Units	Second Semester		Units
ECE C304	Engineering Management	2	EGR C402	ECE Elective 2	4
ECE C303	Feedback and Control	4	ECE C404	ECE Laws, Contracts, Ethics & Standard & Safety	3
ECE C401	ECE Elective 1	4	ECE C406	Design Project 2	1
ECE C403	Environmental Science & Engineering	3	ECE C415	Integrated Course 3	3
ECE C405	Design Project 1	1			
ECE C409	Seminars/Colloquium	2			
ECE C411	Advanced Computer Networks	4			
		20			8

Electives/Track Courses

Track 1:	Artificial Intelligence		8
	Artificial Intelligence 1	4	
	Artificial Intelligence 2	4	
Track 2:	Broadcasting		8
	Broadcast Production Engineering 1	4	
	Broadcast Production Engineering 2	4	

GROUP REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE IN ELECTRONICS ENGINEERING (BSEcE)

	Units	Units
1. Technical Courses		
a. Mathematics		12
Calculus 1	3	
Calculus 2	3	
Engineering Data Analysis	3	
Differential Equations	3	
b. Physical Science		8
Physics for Engineers	4	
Chemistry for Engineers	4	
c. Basic Engineering Science		9
Computer-Aided Drafting	1	
Engineering Economics	3	
Technopreneurship 101	3	
Engineering Management	2	
d. Allied Courses		20
Material Science & Engineering	3	
Physics for ECE	4	
Computer Programming	2	
Environmental Science & Engineering	3	
Circuits 1	4	
Circuits 2	4	
e. Professional Courses		80
Core Courses		

	Electronics Engineering as a Discipline	1	
	Electromagnetics	4	
	Electronic 1: Electronics Devices & Circuit	4	
	Electronic Circuit Analysis & Design	4	
	Advance Engineering Mathematics	4	
	Communications 1: Principles of Communication Systems	4	
	Electronics Shop works	1	
	Digital Electronics 1: Logic Circuits & Switching Theory	4	
	Digital Electronics2: Microprocessor & Microcontroller Systems	4	
	Electronics 3: Electronics Systems & Design	4	
	Communications 2: Modulation & Coding Techniques	4	
	Communications 3: Transmission Media &Antenna System Design	4	
	Signals, Spectra, Signal Processing	4	
	Communications 4: Data Communications	4	
	Internship	3	
	Feedback & Control Systems	4	
	Telephone Systems	3	
	Methods of Research	3	
	ECE Laws, Contracts, Ethics & Standard & Safety	3	
	Design Project 1	1	
	Design Project 2	1	
	Integrated Course 1	1	
	Integrated Course 2	2	
	Integrated Course 3	3	
	Seminars/Colloquium	2	
	Advance Computer Networks	4	
	f. Technical Electives Courses		8
	ECE Elective 1	4	
	ECE Elective 2	4	
2.	Non-Technical Course		
	a. General Education Courses		24
	Art Appreciation	3	
	Science, Technology & Society	3	
	Contemporary World	3	
	Readings in Philippine History	3	
	Understanding the Self	3	
	Purposive Communication 1	3	
	Mathematics in the Modern World	3	
	Ethics	3	
	b. GE Electives/Mandated Courses		12
	Life & Works of Rizal	3	
	Strategies for Academic Success in College	3	

Purposive Communication 2	3	
Linear Algebra	3	
c. Physical Education		8
PATHFIT 1	2	
PATHFIT 2	2	
PATHFIT 3	2	
PATHFIT 4	2	
d. National Service Training Program		6
National Service Training Program 1	3	
National Service Training Program 2	3	
	Total	**187**

Additional Requirements:

1. After their first year, students admitted to the BSEcE Program will be taking a Comprehensive Exam that will cover early Engineering Mathematics courses and science courses, Chemistry, and Physics.

   a. If a student gets the target grade of 60% or above, the student continues with regular status.
   b. If a student fails to achieve the target grade, the student will undergo probationary status.
      i. Students under probationary status will be limited to enroll only 70% of maximum allowable units to be enrolled.
      ii. To lift status, the student needs to get a grade of not lower than 2.5 in all major courses enrolled and without failure. Otherwise, the student will be advised to shift to CPE or any IT-related programs.

2. Students admitted to the BSEcE Program will need to get a grade of 2.5 or better in all their major courses starting in their second year. Students who fail to meet this minimum grade requirement have to re-enroll in the same course until the minimum grade requirement is met and only be allowed to retake the course a maximum of three times. Otherwise, they will be disqualified from the program and be advised to shift to other Engineering programs or IT-related programs.

COMPUTER STUDIES AND ENGINEERING

DEPARTMENT OF COMPUTER ENGINEERING

EGR C101. Computer-Aided Drafting. Concepts of computer-aided drafting (CAD); introduction to the CAD environment; terminologies; and the general operating procedures and techniques in entering and executing basic CAD commands. *Laboratory: 1 unit. Credit: 1 unit. Prerequisite: BSCPE - MAT C602, NSC C104; BSECE - None*.

EGR C103. Engineering Data and Analysis. This course is designed for undergraduate engineering students with an emphasis on problem-solving related to societal issues that engineers and scientists are called upon to solve. It introduces different methods of data collection and the suitability of using a particular method for a given situation. The relationship of probability to statistics is also discussed, providing students with the tools they need to understand how "chance" plays a role in statistical analysis. Probability distributions of random variables and their uses are also considered, along with a discussion of linear functions of random variables within the context of their application to data analysis and inference. The course also includes estimation techniques for unknown parameters; and hypothesis testing used in making inferences from the sample to population, inference for regression parameters, and build models for estimating means and predicting future values of key variables under study. Finally, statistically-based experimental design techniques and analysis of outcomes of experiments are discussed with the aid of statistical software. *Lecture: 3 units. Credit: 3 units. Prerequisite: BSCPE - MAT C601 (STEM), EGR C105 (NON-STEM); BSECE - NONE*.

EGR C105. Mathematics for Engineering. This serves as a bridging course for courses missed by non-STEM freshmen. The course emphasizes the mastery of critical skills and exposure to new skills necessary for success in subsequent math courses. Topics include fundamental concepts of Algebra, functions, and graphs, polynomials and rational functions, exponential and logarithmic functions, trigonometric functions, analytic trigonometry, topics in trigonometry, systems of equations and inequalities, matrices and determinants, conic sections and analytic geometry, sequences, induction, probability, and an introduction to Calculus. *Lecture: 5 units. Credit: 5 units. Prerequisite: None (Non-STEM Track).*

EGR C106. Mathematics for Engineering 1. This course provides a comprehensive foundation in mathematics tailored for engineering students. It encompasses three primary areas: basic mathematics, algebra, and advanced algebra. Each section is designed to equip students with the mathematical tools and techniques essential for solving engineering problems. *Lecture: 3 units. Laboratory: 0 unit. Credit: 3 units. Prerequisite: None.*

EGR C107. Mathematics for Engineering 2. This course is a continuation of the foundational mathematics necessary for engineering students, focusing on three key areas: trigonometry, solid mensuration, and analytic geometry. Each section aims to deepen students' understanding of mathematical concepts and enhance their ability to apply these concepts in engineering contexts. *Lecture: 3 units. Laboratory: 0 unit. Credit: 3 units. Prerequisite: None.*

EGR C201. Differential Equations. This course is intended for all engineering students to have a firm foundation on differential equations in preparation for their degree-specific advanced mathematics courses. It covers first-order differential equations, nth order linear differential equations, and systems of first-order linear differential equations. It also introduces the concept of Laplace Transforms in solving differential equations. The students are expected to be able to recognize different kinds of differential equations, determine the existence and uniqueness of the solution, select the appropriate methods of solution, and interpret the obtained solution. Students are also expected to relate differential equations to various practical engineering and scientific problems as well as employ computer technology in solving and verifying solutions. *Lecture: 3 units. Credit: 3 units. Prerequisite: MAT C602.*

EGR C203. Engineering Economics. This course deals with the study of concepts of the time value of money and equivalence, basic economic study methods; decisions under certainty; decisions recognizing risk; and decisions admitting uncertainty. *Lecture: 3 units. Credit: 3 units. Prerequisite: BSCPE - MAT C602, NSC C104; BSECE - EGR C103.*

EGR C204. Technopreneurship 101. This introductory course provides learning on the fundamental concepts and processes related to starting and managing technology-based new ventures. It encourages the students to develop entrepreneurial mindset, creating thinking to drive innovative ideas of technology-based solutions that solve local and global needs. Students will have a clear understanding of how to evaluate market opportunities, develop a business model, raise capital, design and develop a minimum viable product as an entrepreneurial venture for startups or R&D product that industry currently caters. During the course, students will develop their own business plan and can exhibit their solutions to potential stakeholders. *Lecture: 3 units. Credit: 3 units. Prerequisite: BSCPE - CPE C307, CPE C206; BSECE - ECE C405.*

EGR C205. Circuits 1. This course introduces the fundamental concepts, circuit laws, theorems, and techniques used in electrical circuit analysis and transient analysis, as well as its application. The course covers circuit topologies and DC excitations, transient response, AC response, and polyphase circuits. The use of computer software for circuit simulation and design is emphasized to expose students to computer-based tools. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: BSCPE - NSC C104; BSECE – ECE C104.*

EGR C206. Circuit 2. Complex algebra and phasors; simple AC circuits, impedance, and admittance; mesh and node analysis for AC circuits; AC network theorems; power in AC circuits; resonance; three-phase circuits; transformers; two-port network parameters and transfer function. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: EGR C205.*

EGR C302. Methods of Research. This course will provide an in-depth understanding of research through the exploration of different research methodologies and ethics. It includes qualitative and quantitative research, descriptive and other applicable research methodologies, inferential statistics, and introduction to data mining. *Lecture: 3 units. Credit: 3 units. Prerequisite: CPE C306, EGR C103, ENG C102.*

CPE C101. Computer Engineering as a Discipline. This course discusses the curriculum for Computer Engineering and how to prepare students for success through the engineering design process, ethical decision-making, teamwork, and communicating to diverse audiences. *Lecture: 1 unit. Credit: 1 unit. Prerequisite: NONE.*

CPE C102. Object-Oriented Programming for CPE. This course introduces the fundamental concepts of programming from an object-oriented perspective. Topics are drawn from classes and objects, abstraction, encapsulation, data types, calling methods and passing parameters, decisions, loops, arrays and

collections, documentation, testing and debugging, exceptions, design issues, inheritance, and polymorphic variables and methods. The course emphasizes modern software engineering and design principles. *Laboratory: 2 units. Credit: 2 units. Prerequisite: CPE C103.*

CPE C103. Programming Logic and Design. This is an introductory course in computer programming logic. The student will learn algorithms applicable to all programming languages, including identifiers, data types, arrays, control structures, modular programming, generating reports, and computer memory concepts. The student will learn to use charts commonly used in business and information processing. Program logic will be developed using flowcharts and pseudocode. Programs will be written using any programming language. *Laboratory: 2 units. Credit: 2 units. Prerequisite: NONE.*

CPE C201. Data Structures and Algorithms. Solving computational problems that involve manipulating collections of data, study a core set of data abstractions, data structures, and algorithms that provide a foundation for writing efficient programs. *Laboratory: 2 units. Credit: 2 units. Prerequisite: CPE C102.*

CPE C202. Software Design. The course discusses the principles, techniques, and tools used to effect the orderly production of medium and large scale computer software. It includes problem-solving concepts, software development process, software requirements, and specifications, verification, and validation. These techniques will be applied to programming projects with students working in teams and managing all phases of programming projects using Python as the development tool. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units Prerequisite: CPE C201.*

CPE C203. Computer Engineering Drafting and Design. This course focuses on the principles of the layout of electrical, electronic, and logic drawings, stressing modern representation used for block diagrams, wiring/assembly, drawings, printed circuit board layouts, and etching. *Laboratory: 1 unit. Credit: 1 unit. Prerequisite: ECE C203.*

CPE C204. Computer Systems and Troubleshooting. This course will provide an introduction to microcomputer system hardware, operating systems, and application software. Installation of basic Local Area Network (LAN) is also included. It covers topics on microcomputer installation, servicing, and troubleshooting techniques. Concepts are learned through extensive hands-on activities. The proper use and care of tools and equipment are emphasized in this course. *Lecture: 2 units. Laboratory: 1 unit. Credit: 3 units. Prerequisite: EGR C205.*

CPE C205. Fundamentals of Mixed Signals and Sensors. This course covers operational amplifiers, signal converters, power-switching devices, and the construction and operation of sensors and transducers for converting physical parameters into electrical signals and vice-versa. The course focuses on the application of these devices in developing signal conversion circuits that allow measurement, processing, and control of physical parameters by digital processing systems such as a finite state machine or a digital computer. Topics on actuators are also included. *Lecture: 3 units. C redit: 3 units. Prerequisite: ECE C203*.

CPE C206. Feedback and Control Systems. The course includes the control devices, equations of a system, and block diagrams of systems. *Lecture: 3 units. Credit: 3 units. Prerequisite: MAT C603, EGR C205*.

CPE C301. Logic Circuits and Design. The course includes the design and analysis of digital circuits. This course covers both combinational (synchronous and asynchronous) logic circuits with emphasis on solving digital problems using hardwired structures of the complexity of medium and large-scale integration. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C203*.

CPE C302. Computer Networks and Security. The course includes the basic principles of network architecture, computer network design, services, technologies, and network security. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: CPE C303*.

CPE C303. Data and Digital Communications. This course focuses on the fundamental concepts of digital and data communications. It also includes topics on data security and integrity. *Lecture: 3 units. Credit: 3 units. Prerequisite: ECE C203*.

CPE C304. Basic Occupational Health and Safety. This course tackles key Occupational Health and Safety (OSH) concepts, principles, and practices that are foundational knowledge requirements applicable in almost all industries. Specifically, it assists learners in identifying the key elements in the OSH situation both here and abroad; determine existing and potential safety and health hazards; identify the range of control measures; discuss pertinent provisions of Philippine laws that refer to occupational safety and health; explain key principles in effectively communicating OSH; identify components of effective OSH programs and demonstrate some skills in identifying hazards and corresponding control measures at the workplace. *Lecture: 3 units. Credit: 3 units. Prerequisite: CPE C307, CPE C206*.

CPE C305. Digital Signal Processing. The course includes the need for and tradeoffs made when sampling and quantizing a signal; linear, time-invariant system properties; frequency as an analysis domain complementary to time; and

filter design. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: CPE C206.*

CPE C306. Microprocessors. This course provides an understanding of the architecture of microprocessor-based systems, registers, study of microprocessor operation, assembly language, arithmetic operations, and interfacing. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: CPE C301.*

CPE C307. Cognate/Track Course 1. Advanced Database Management (Software Development Tack). This advanced course explores database management systems and associated technologies, the management of database systems, and the data modeling options facilitating the storage and use of data. An investigation into the theories that govern the design of database systems is combined with an analysis of advanced concepts through investigation and subsequent practical implementation and critical evaluation. *Lecture: 2 units. Laboratory: 1 unit. Credit: 3 units. Prerequisite: ECE C203, CPE C315, MAT C603.*

CPE C307. Cognate/Track Course 1. Network and System Administration (System and Network Administration Track). This course explores the basic operation and management of local and wide area networks using the network operating system (NOS). Topics include installation of server and work-station software, physical network configuration, network security, policy, domain controllers, performance monitoring, and troubleshooting techniques. NOS features, ease of management, utilities, upgrades, and interoperability with other NOSs and client types are analyzed. *Lecture: 2 units. Laboratory: 1 unit.Credit: 3 units. Prerequisite: ECE C203, CPE C315, MAT C603.*

CPE C307. Cognate/Track Course 1. Online Technology. This course presents the Internet from a dynamic workplace perspective (Information Technology Track). It reflects on how emerging technologies will empower society to do more with the Internet. This course covers core Internet technologies, Web page design and authoring, computational thinking, networking fundamentals, and technology planning. *Lecture: 2 units. Laboratory: 1 unit. Credit: 3 units. Prerequisite: ECE C203, CPE C315, MAT C603.*

CPE C308. Cognate/Track Course 2. Application Development (Software Development Tack). This course will introduce students to developing applications that target web and mobile devices. Students will be introduced to many issues unique to web and mobile applications, including synchronization, remote data access, and security. *Lecture: 2 units. Laboratory: 1 unit. Credit: 3 units. Prerequisite: Cognate/Track Course 1.*

CPE C308. Cognate/Track Course 2. Network Security (System and Network Administration Track). This course introduces students to network security. By examining case studies and reading seminal research papers, students will learn about network attacks and vulnerabilities as well as current defenses. Topics covered include cryptography, confidentiality, authentication protocols, botnets, firewalls, intrusion detection systems, and communication privacy and anonymity. *Lecture: 2 units. Laboratory: 1 unit. Credit: 3 units. Prerequisite: Cognate/Track Course 1.*

CPE C308. Cognate/Track Course 2. Management and Information System (Information Technology Track). This course will focus on what MIS is, how they influence your current or prospective jobs, why they impose specific - and sometimes seemingly absurd - operational procedures, and how to use this knowledge to your advantage in your professional life. This MIS course will cover supporting tech infrastructures (Cloud, Databases, Big Data), the MIS development/ procurement process, and the main integrated systems. *Lecture: 2 units. Laboratory: 1 unit. Credit: 3 units. Prerequisite: Cognate/Track Course 1.*

CPE C309. Introduction to HDL. A laboratory course that introduces hardware description language as a tool for designing and testing combinational and sequential circuits. It covers fundamental concepts of HDL and the basic building blocks of HDL programming. *Laboratory: 1 unit. Credit: 1 unit. Prerequisite: CPE C103, ECE C203.*

CPE C310. CpE Laws and Professional Practice. This course provides the importance of the professional and ethical responsibilities of practicing computer engineers and the effects of their work on society, the importance of understanding contemporary issues, lifelong learning strategies, and applicable IT laws in the field of computer engineering. *Lecture: 2 units. Credit: 2 units. Prerequisite: ECE C203, CPE C315, MAT C603.*

CPE C311. Operating Systems. This course includes different policies and strategies used by an operating system. Topics include operating systems structures, process management, and storage management, file management, and distributed systems. *Lecture: 3 units. Credit: 3 units. Prerequisite: CPE C201.*

CPE C312. Embedded Systems. This course provides advanced topics in embedded systems design using contemporary practice, interrupt-driven, reactive, real-time, object-oriented, and distributed client/server embedded systems. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: CPE C306.*

CPE C313. Artificial intelligence, 1. This course provides introductory concepts of Artificial Intelligence. It exposes students to the basic ideas, challenges, techniques, and problems in artificial intelligence. It will cover the history, theory and computational methods of artificial intelligence. Basic Concepts include representation of knowledge and computational methods for reasoning. One or two application areas will be studied, to be selected from expert systems, robotics, computer vision, natural language understanding, and planning. *Lecture: 2 units. Laboratory: 1 unit. Credit: 3 units. Prerequisite: MAT C202, CPE C315, CPE C301.*

CPE C314. Artificial intelligence 2. This course is the second part of a course series in Artificial Intelligence. It will cover more topics on intelligent systems in AI, machine learning algorithms for supervised and unsupervised learning, neural networks and natural language processing. Students will be required to work on real-world problems and proposed potential solutions applying to any of the domains of AI. *Lecture: 3 units. Laboratory: 1 unit. Credit: 3 units. Prerequisite: CPE C313.*

CPE C315. Computer Application. This course will focus on the utilization of the current computer program/software to perform a specific useful task. Students will have practical and hands-on experience in computer programs or software, which can be used in various applications. This also focuses on the management and use of new technologies (past, present, and future), more especially on the effective integration of information technologies (e.g., programming language, software packages, operating systems, security issues, and hardware infrastructure) within organizations. During the course, each student will develop a broad understanding of new technologies in general while developing their knowledge/skill in the use of one particular technology. Learning will be accomplished through assigned reading, class discussion, and hands-on use of state-of-the-art technology. *Laboratory: 1 unit. Credit: 1 unit. Prerequisite: CPE C201, ITC C201.*

CPE C401. Computer Architecture and Organization. This course includes the study of the evolution of computer architecture and the factors influencing the design of hardware and software elements of computer systems. The focus is on the understanding of the design issues; specifically, instruction set architecture and hardware architecture. *Lecture:3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: CPE C306.*

CPE C402. Internship. This course enables students to relate their acquired competencies to the realities and problems of industries in a multidisciplinary environment. This may include involvement in the industry's manpower requirements, development and research concerns, training, applications of principles, environmental concerns, ethical and behavioral concerns, decision-making, and equipment and materials concerns. *Lecture:3 units. Credit: 3 units. Prerequisite: CPE C403, CPE C407, CPE C409 (required 240 hrs.).*

CPE C403. CpE Practice and Design 1. This course is the first course in a two-semester sequence that constitutes the design experience for undergraduate computer engineers. It provides essential ideas, concepts, and principles in the engineering design process and emphasizes other design issues, including engineering standards and multiple constraints, as well as effective communication strategies. Students work in teams to develop project proposals for assigned open-ended problems. Students are required to make oral presentations and submit a written proposal for their projects. *Laboratory: 1 unit. Credit: 1 unit. Prerequisite: CPE C306, EGR C302.*

CPE C404. CpE Practice and Design 2. This course is the second of the design experience for undergraduate computer engineering students. In this course, students will be expected to build/fabricate their design, test and evaluate the design against their design specifications, and demonstrate a fully functional project to their design review committee. Students make oral presentations and submit final reports documenting their projects. *Laboratory: 2 units. Credit: 2 units. Prerequisite: CPE C403.*

CPE C405. Emerging Technologies in CpE. This 3-unit course focuses on emerging technologies in the field of computer engineering, IT trends and other allied disciplines that would be relevant for the computer engineering profession. This course will specifically discuss the core concepts of cloud computing and expose students to popular cloud computing platform like AWS. It will cover topics on AWS cloud computing model, architecture, services, products, pricing, technology, security, and shared responsibility model. The course will prepare students for Cloud Practitioner certification. *Lecture: 3 units. Credit: 3 units. Prerequisite: EGR C302.*

CPE C407. Cognate/Track Course 3. Data Science (Software Development Tack). Data Science is the study of the generalized extraction of knowledge from data. It requires an integrated skill set spanning mathematics, statistics, machine learning, databases, and other branches of computer science, along with a good understanding of the craft of problem formulation to engineer effective solutions. This course will introduce students to this rapidly growing field and equip them with some of its basic principles and tools, as well as its general mindset. Students will learn concepts, techniques, and tools they need to deal with various facets of data science practice, including data collection and integration, exploratory data analysis, predictive modeling, descriptive modeling, data product creation, evaluation, and effective communication. The focus in the treatment of these topics will be on breadth rather than depth, and emphasis will be placed on the integration and synthesis of concepts and their application to solving problems. Real datasets from a variety of disciplines will be used to make the learning contextual. *Lecture:2 units. Laboratory: 1unit. Credit: 3 units. Prerequisite: Cognate/Track Course 2.*

CPE C407. Cognate/Track Course 3. Cyber Security (System and Network Administration Track). This course introduces students to the interdisciplinary field of cybersecurity by discussing the evolution of information security into cybersecurity, cybersecurity theory, and the relationship of cybersecurity to nations, businesses, society, and people. Students will be exposed to multiple cybersecurity technologies, processes, and procedures, learn how to analyze the threats, vulnerabilities, and risks present in these environments, and develop appropriate strategies to mitigate potential cybersecurity problems. *Lecture: 2 units. Laboratory: 1unit. Credit: 3 units. Prerequisite: Cognate/Track Course 2.*

CPE C407. Cognate/Track Course 3. Emerging Technology (Information Technology Track). Technology is rapidly changing and evolving. An IT professional must be able to identify the potential benefits of new technology and determine the feasibility of implementation into a given system. Students will learn to research and apply new and/or innovative technologies that are being integrated into the Information Technology environment. This course will provide students with the opportunity to investigate trends and examine the potential impact of the technology. *Lecture: 2 units. Laboratory: 1 unit, Credit: 3 units. Prerequisite: Cognate/Track Course 2.*

CPE C409. Seminars and Field Trips. The course includes seminars and lectures on current trends and issues in Computer Engineering developments. Include field trips to different companies and plants dealing with computer system facilities. *Laboratory: 2 units. Credit: 2 units. Prerequisite: EGR C302.*

DEPARTMENT OF ELECTRONICS ENGINEERING

ECE C101. Electronics Engineering as a Discipline. Electronics Engineering, as a Discipline, assists new students in what to expect in the degree. This includes familiarization of the curriculum, their professional organization, and their possible careers in the future. *Lecture:1 unit. Credit: 1 unit. Prerequisite: NONE.*

ECE C102. Material Science and Engineering. This course introduces the students to a broad study of the structure and composition of materials (metals, polymers, ceramics, and composite materials) and their properties and behavior in service environments. *Lecture: 3 units. Credit: 3 units. Prerequisite: NSC C204.*

ECE C104. Physics for ECE (Physics 2). Thermodynamics (1st & 2nd Law, basic concepts on heat engine and refrigerators), Energy Conversion (EM Induction, magnetic flux, generators), Semiconductor Physics. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: MAT C601, NSC C104 (co-requisite).*

ECE C201. Computer Programming. This course introduces the fundamental concepts of programming from an object-oriented perspective. Topics are drawn from classes and objects, abstraction, encapsulation, data types, calling methods and passing parameters, decisions, loops, arrays and collections, documentation, testing and debugging, exceptions, design issues, inheritance, and polymorphic variables and methods. The course emphasizes modern software engineering and design principles. *Laboratory: 2 units. Credit: 2 units. Prerequisite: NONE*.

ECE C202. Electromagnetics. This course deals with vector algebra, vector calculus, vector analysis, and their applications in electric and magnetic fields, resistive, dielectric, and magnetic materials, coupled circuits, magnetic circuits and fields, time-varying electromagnetic fields, and Maxwell's equations. *Lecture: 4 units. Credit: 4 units. Prerequisite: EGR C201*.

ECE C203. Electronics 1: Electronic Devices and Circuits. This course discusses the construction, operation, and characteristics of basic electronic devices such as junction diodes, bipolar junction transistors, Field Effect Transistors, and MOS Field-Effect Transistors and oscillators. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: BSCPE - EGR C205; BSECE – ECE C104*.

ECE C204. Electronics 2: Electronics Circuit Analysis and Design. High-frequency transistor models; analysis of transistor circuits; multi-stage amplifier, feedback, differential amplifiers, and operational amplifiers; integrated circuit families (RTL, DTL, TTL, ECL, MOS). *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C203*.

ECE C205. Advanced Engineering Mathematics for ECE. A study of selected topics in mathematics and their applications in advanced courses in engineering and other allied sciences. It covers the study of Complex numbers and complex variables, Laplace and Inverse Laplace Transforms, Power series, Fourier series, Fourier Transforms, z-transforms, power series solutions of ordinary differential equations, partial differential equations, and numerical methods in engineering. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4. Prerequisite: EGR C201*.

ECE C206. Communications 1: Principles of Communications Systems. Bandwidth; filters; linear modulation; angle modulation; phase-locked loop; pulse modulation; multiplexing techniques; noise analysis; radio transmitters and receivers. *Lecture: 3 units, Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C203, ECE C204 (co-requisite)*.

ECE C207. Electronics Shop work. An introduction to the different tools and practices in an electronics shop. This includes the utilization of CAD tools, PCB **development, and the like to assist students in the construction of different projects they** will encounter throughout their stay as Electronics Engineering students. *Laboratory: 1 unit, Credit: 1 unit. Prerequisite: ECE C104.*

ECE C215. Integrated Course 1. This course is specifically designed to prepare engineering students for their board exams by providing a comprehensive review of key mathematical concepts and problem-solving techniques. It focuses on reinforcing students' understanding and proficiency in mathematics, ensuring they are well-equipped to excel in their board exams. *Lecture: 0 unit. Lab: 3 units. Credit: 1 unit. Prerequisite: EGR C201.*

ECE C301. Digital Electronics 1: Logic Circuits and Switching Theory. Review of number systems, coding, and Boolean algebra; inputs and outputs; gates and gating networks; combinational circuits; standard form; minimization; sequential circuits; state and machine equivalence; asynchronous sequential circuits; race conditions; algorithmic state machines; design of digital subsystems. *Lecture: 3 units, Laboratory: 1 unit. Credit: 4 Units. Prerequisite: ECE C204.*

ECE C302. Digital Electronics 2: Microprocessor and Microcontroller system. The course covers concepts involving microprocessor/microcontroller systems architecture/organization, including microprocessor/microcontroller programming, interfacing techniques, memory systems, and bus standards. In the laboratory, the students will be involved with experiments using microcontrollers and the use of microprocessor/microcontroller development systems and other tools. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite:* ECE C301.

ECE C303. Electronics 3: Electronics System and Design. Theory, operating characteristics, and design of electronic devices and control circuits for industrial processes; industrial control applications; electronics instrumentation; transducers; data acquisition system; interfacing techniques; sensors. *Lecture: 3 units, Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C204.*

ECE C304. Engineering Management. This course will entail students to learn the basic function of a manager applicable in decision making, which applies to real-world problems. Furthermore, students would learn how to apply planning, leading, organizing, and control principles into the resources to increase efficiency. *Lecture: 2 units. Credit: 2 units. Prerequisite: EGR C103.*

ECE C305. Communications 2: Modulation and Coding Techniques. Random variables, bit error rate; matched filter; Digital modulation techniques;

ASK, FSK, QAM, PSK/QPSK, CDMA, and W-CDMA systems; signal space; generalized orthonormal signals; information measures-entropy; channel capacity; efficient encoding; error-correcting codes information theory; data compression; coding theory. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C206*.

ECE C306. Communications 3: Transmission Media and Antenna System Design. Transmission media; radio wave propagation wire and cable transmission systems; fiber-optic transmission system; transmission lines and antenna systems. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C305*.

ECE C307. Signals, Spectra, Signal Analysis. Fourier transform; z transform; convolution; FIR filters; IIR filters; random signal analysis; correlation functions; DFT; FFT; spectral analysis; applications of signal processing to speech, image, etc. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C205*.

ECE C308. Communications 4: Data Communications. Data communications systems; terminals, modems; terminal control units; multiplexers; concentrators; front-end processors; common carrier services; data communication system design; computer network models; TCP/IP; principles; LAN; WAN. *Lecture: 3 units, Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C305*.

ECE C309. Internship. Actual On-the-Job Training or Industry Internship in the field of specialization. Hours to be rendered: 240 hours. *Lecture: 3 units. Credit: 3 units. Prerequisite: ECE C308; ECE C302*.

ECE C310. Feedback and Control System. This course deals with the time and frequency response of feedback control systems. The topics covered include time response of first-order and second-order systems, modeling, transfer functions, pole-zero map, stability analysis, root locus, bode plots, compensators, PID controllers, and introduction to state-space techniques. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C205*.

ECE C312. Telephone Systems. Operating performance and interface standards for voice and data circuits; telecommunications facility planning; outside plant engineering; surveying; switching and handling systems; mobile systems and standards; cellular radio systems; PSTN. *Lecture: 3 units. Credit: 3 units. Prerequisite: ECE C305*.

ECE C314. Methods of Research. This course deals with research preparation methods, research tools, research proposals, and the implementation,

presentation, and publication of research work. *Lecture: 3 units. Credit: 3 units. Prerequisite: ECE C303*

ECE C315. Integrated Course 2. This course is designed to provide an intensive review for engineering students preparing for their board exams, with a dual focus on advanced mathematics and a thorough review of general engineering and applied science concepts. The course aims to solidify students' foundational knowledge and enhance their problem-solving skills across multiple disciplines. *Lecture: 0 unit. Lab: 6 units. Credit: 2 units. Prerequisite: ECE C215*

ECE C401. ECE Elective 1 (Broadcast Production Engineering 1). Discusses operation of audio and video equipment including amplifiers, processors, audio/video mixers, distribution amps, TV cameras, microphones, monitors systems integration, studio electro-acoustic and lighting, TV and radio transmitters and propagation, coverage map calculation and frequency analysis, broadcast networking, broadcast ancillary services (STL's and satellite links). It also includes CATV technology and DTH. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C306, ECE C308.*

ECE C402. ECE Elective 2 (Broadcast Production Engineering 2). The course includes the applications in different areas of broadcasting such as television, AM, FM, cable television, telecommunications, data communications, studio acoustics, etc. through experiments and field research; basic equipment or devices used for transmission of signals such as filters and oscillators, radio frequency power amplifiers and mixers, basic circuits of modulation and demodulation, transmitters and studio equipment. *Lecture: 3 units. Laboratory: 1 unit. Credit: 4 units. Prerequisite: ECE C401*.

ECE C403. Environmental Science and Engineering. Environmental Science Knowledge in Ecology and Human Population Control, Variety of Resources and Outline Plans for Attaining Sustainable Society, The Enigma of Pollution, and the Legal, Technical and Personal Solutions for it. Study of Environmental Impact Assessment and Environmental Crisis. *Lecture: 3 units. Credit: 3 units. Prerequisite: NSC C204*.

ECE C404. ECE Laws, Contracts, Ethics, and Standards and Safety. Contracts; warranties; liabilities; patents; bids; insurance; other topics on the legal and ethical positions of the professional engineer. This course includes Safety and other standards related to the ECE profession. *Lecture: 3 units. Credit: 3. Prerequisite: NONE*.

ECE C405. Design Project 1. This capstone course utilizes the fundamentals of electronics engineering in the design of an electronic system. It includes the synthesis of processes, analysis of process conditions, and the analytic, heuristic, and optimum design of equipment and processes. Economic analysis is included to estimate the cost of equipment, capital investment, total product cost, and profitability. *Lecture: 1 unit. Credit: 1 unit. Prerequisite: ECE C314, ECE C308, ECE C302.*

ECE C406. Design Project 2. This capstone course utilizes the fundamentals of electronics engineering in the design of an electronic system. It includes the synthesis of processes, analysis of process conditions, and the analytic, heuristic, and optimum design of equipment and processes. Economic analysis is included to estimate the cost of equipment, capital investment, total product cost, and profitability. *Laboratory: 1 unit, Credit: 1. Prerequisite: ECE C405.*

ECE C407. Integrated Course 1. This course is the recapitulation and synthesis of the various courses of Electronics Engineering taken during the lower years. Pre-board reviews and qualifying examinations, particularly in general engineering courses and mathematics. *Lecture: 3 units. Credit: 3 units. Prerequisite: ECE C303; ECE C308.*

ECE C408. Integrated Course 2.This course is the recapitulation and synthesis of the various courses of Electronics Engineering taken during the lower years. Pre-board reviews and qualifying examinations, particularly in electronics and electronics systems technology. *Lecture: 3 units. Credit: 3 units. Prerequisite: ECE C407.*

ECE C409. Seminars/Colloquium. This course deals with a series of lectures and seminars on selected topics that are highly relevant to electronics engineering but are not covered in any of the other formal courses. It covers recent advances in electronics engineering. It is also a venue for the students to present their projects and researches. *Lecture: 2 units. Credit: 2 units. Prerequisite: ECE C302, ECE C308, ECE C314.*

ECE C411. Advanced Computer Networks. The course includes the basic principles of network architecture, computer network design, services, technologies, and network security. *Credit: 4 units. Prerequisite: ECE C308.*

ECE C415. Integrated Course 3. This course provides an all-encompassing review for engineering students preparing for their board exams, covering mathematics, general engineering, applied science, and specialized topics in electronics and communication. It aims to equip students with the knowledge and

skills necessary to excel in all areas tested in the board exams. *Lecture: 0 unit. Lab: 9 units. Credit: 3 units. Prerequisite: ECE C315*